



Daffa Naufal Mumtaz Heryadi

Bekasi, Indonesia

+62 896-6358-0475

— naufalmumtazdaffa037@gmail.com

— [LinkedIn](#)

[GitHub](#)

— [Portfolio](#)

Computer Science student at IPB University focused on **software development**, **data science**, and **applied AI/ML**. Experienced in building **practical applications**, supporting **data-driven decision-making**, and working with **Python-based machine learning workflows**. Currently seeking internship opportunities in software development, data science, or AI engineering, particularly in tech and fintech environments.

Education & Training

IPB University - Bogor, Indonesia

2023 – 2027 (Expected)

Bachelor of Computer Science

GPA: **3.81/4.00**

Relevant Coursework: Data Structures, Databases, Statistics and Data Analysis, Artificial Intelligence, Data Mining, Software Engineering, System Analysis and Design, User Experience Design

ASAH 2025 Machine Learning Cohort by Dicoding - Remote

2025

Machine Learning Cohort

- Completed **450** learning hours in **machine learning**, **Python-based workflows**, model development, and applied AI/ML projects; completed the **Sigma-D** capstone with a final score of **88.28**.

Experience

Teaching Assistant - Department of Computer Science, IPB University

Aug 2025 – Jun 2026

Bogor, Indonesia

Introduction to Computational Mathematics, Aug–Dec 2025 — User Experience Design, Jan–Jun 2026

- Assisted **80+** undergraduate students across two courses over two semesters: Introduction to Computational Mathematics and User Experience Design.
- Supported **28** practicum sessions by guiding students through exercises, assignments, and project-related discussions.
- Reviewed and provided feedback on **20+** assignments, quizzes, and projects while helping maintain grading records and learning materials.

Organizational Experience

Himpunan Mahasiswa Ilmu Komputer (Himalkom) - IPB University

Jan 2025 – Dec 2025

Staff of Research and Technology

- Coordinated the planning, execution, and analysis of a survey project assessing students' interests and skills, gathering **100+** responses and collaborating with representatives from other divisions.
- Contributed to data analysis and simple data visualization for surveys involving **120–160** respondents, working closely with division members to analyze findings effectively.

IEEE Student Branch IPB - IPB University

Jan 2024 – Dec 2025

Administration, 2024 — Treasurer, 2025

- Managed administrative records and documentation for **24+** programs, including meeting minutes, correspondence, and member data.
- Maintained member and committee databases for **61+** members to support internal coordination.
- Tracked expenses and transaction records, managing up to **IDR 7 million** in organizational funds to support transparent budgeting and evaluation.

Selected Projects

TitikTemu Lost and Found IPB

Software Engineering, Jan – Jun 2026

Front-End and UX Designer — Team of 4

[GitHub](#) — [Live Demo](#)

- Developed frontend and UX flows for a lost-and-found web platform with **16** navigation views, **3** user roles (user, admin, security), and integration with **26** FastAPI endpoints and **7** PostgreSQL tables.
- Designed end-to-end workflows for item reporting, moderation, claim processing, notifications, and security hand-over to support structured lost-and-found operations.

PortalRancajaya Village Profile CMS

Full-Stack Web Development, Jun – Aug 2026

Solo Full-Stack Web Developer and UX/Content System Designer

[GitHub](#) — [Live Website](#)

- Built and deployed a village information portal using **Astro** and **Decap CMS**, structuring **61** content/data files across **13** CMS collections.
- Organized content for public services, village profiles, UMKM, agriculture, APBDes, news, and administrative information to support maintainable village data publishing.

Customer Segmentation MLOps

AI/ML and MLOps, Sep – Dec 2025

Solo Machine Learning and MLOps Developer

[GitHub](#)

- Built an end-to-end local **MLOps** pipeline for customer segmentation, including preprocessing automation, K-Means experiment tracking with **MLflow**/DagsHub artifacts, Flask model inference, Dockerized components, and Prometheus/Grafana monitoring.
- Evaluated **7** clustering configurations and selected $k = 2$ with a **0.5164** silhouette score, while exposing inference and system metrics through **10** Prometheus metric families.

Sigma-D: Predictive Maintenance Copilot

AI/ML, MLOps, Full-Stack AI Application, Oct – Dec 2025

Machine Learning Engineer — Team of 5

[GitHub](#) [Organization](#) — [ML Repo](#) — [Demo](#)

- Developed the ML component for Sigma-D, a predictive maintenance capstone project using **10,000** machine sensor records for binary failure prediction and **6**-class failure-type classification.
- Prepared model outputs, technical documentation, and presentation materials to support integration with the project's AI-assisted maintenance dashboard and ASAH 2025 capstone submission.

MyToko MSME E-Commerce SaaS

Full-Stack Web Development, Sep – Nov 2025

Frontend Developer, UI Designer, Supabase Contributor — Team of 3

[GitHub](#)

- Collaborated in a **3**-person team to build MyToko, a SaaS-style e-commerce platform for MSMEs using **Next.js** and **Supabase**, covering storefront, checkout, invoice, order tracking, wishlist, reviews, and admin management workflows.
- Contributed to frontend/UI development, Supabase integration, technical documentation, and final product presentation for a deployed prototype.

Additional Projects

AQI Bangladesh Forecasting

Data Science / Time-Series Forecasting, 2026

[GitHub](#) Built a LightGBM AQI forecasting pipeline using **1,048,551** records from **30** cities, achieving MAE **1.13**, RMSE **1.77**, and R^2 **0.9986**.

Produce Freshness Classifier

Computer Vision / Image Classification, 2026

[GitHub](#) Compared SVM, Random Forest, and MobileNetV2 across **11** scenarios, achieving **98.70%** accuracy and weighted F1-score on **4,391** test images.

PSO TSP Optimizer - West Java Route Optimization

Computational Intelligence / Optimization, 2026

[GitHub](#) Built a Discrete PSO optimizer and Streamlit dashboard for **27** West Java locations, improving route distance from **891.94 km** to **784.61 km** (**12.03%** improvement).

CookHub: AI-Assisted Cooking Companion

UX Research / Product Design, 2025

[Prototype](#) — [Research Board](#) Designed UX research artifacts and an interactive Figma prototype for AI-assisted recipe discovery.

Certifications

Associate Data Scientist, Badan Nasional Sertifikasi Profesi (BNSP) - LSP TIK Global

Jul 2026

[Credential Link](#)

Achievements

- Best Graduate, Trenmatika Islamic Boarding High School, Bekasi.
 - 2nd Place, AgriTechJam - Agriinformatics.
- 20232025

Skills

Hard Skills: Software development, data science, AI/ML, frontend/backend development, database design, MLOps, data visualization, model evaluation, UX research, product design

Programming Languages: Python, JavaScript/TypeScript, SQL, Java, C/C++, Scala

Frameworks: React, Astro, FastAPI, Tailwind CSS, Vite

Data/ML: TensorFlow, PyTorch, Scikit-Learn, Hugging Face, MLflow, data visualization, model evaluation

Tools: Git, GitHub, GitHub Actions, Docker, PostgreSQL, Supabase, Firebase, Prometheus, Grafana, Figma

Soft Skills: Problem solving, collaboration, technical communication, continuous learning, adaptability, analytical thinking

Languages: Indonesian (Native), English (Professional Working Proficiency), Japanese (Basic), Arabic (Basic)